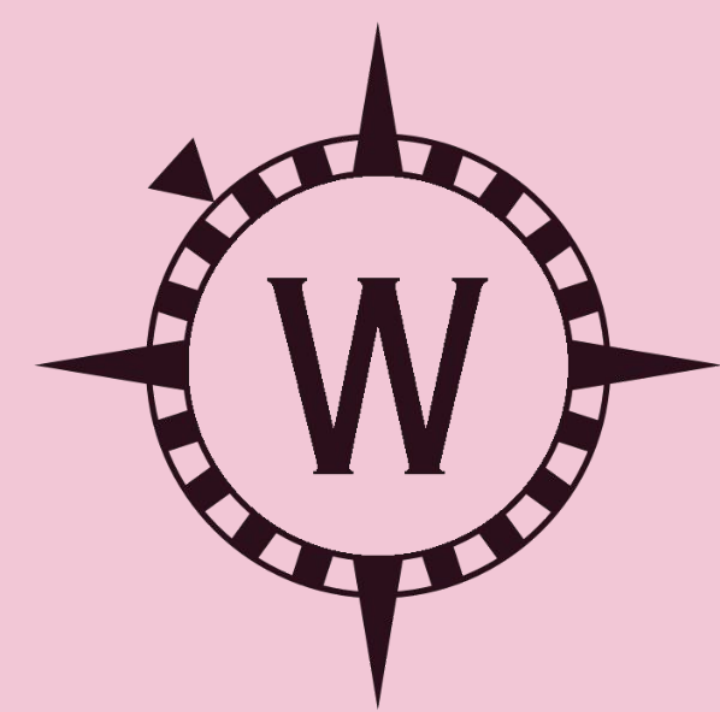




HEARD BUT NOT STUDIED

Mapping the gap between **health research** and women’s **lived experiences**.

Research Question

What women’s health **symptoms** and **conditions** are most discussed on **r/WomensHealth**, and to what extent are these topics represented in clinical research on **PubMed** and **ClinicalTrials.gov**?

Background & Related Work

Previous studies used funding patterns and participant data to assess gaps in women’s health research. They found that:

- **Men’s health research gets 1.84x more funding**
A 2021 study looked at the funding of research towards male-dominant vs female dominant diseases by the U.S. National Institutes of Health (NIH). They found that in 2019, the NIH funded research for men’s diseases 1.84 times more than women’s.^[3]
- **7 out of 11 research areas underrepresented women**
A 2019 study quantified sex bias in clinical studies by analyzing participant data of common diseases that affect men and women. They found substantial underrepresentation of women in 7 of the 11 disease categories they analyzed.^[4]
- **Endometriosis** is estimated to affect around **10% of women worldwide**, yet diagnosis can take up to **10 years**, in part because endometriosis is frequently dismissed as psychological.^[5]

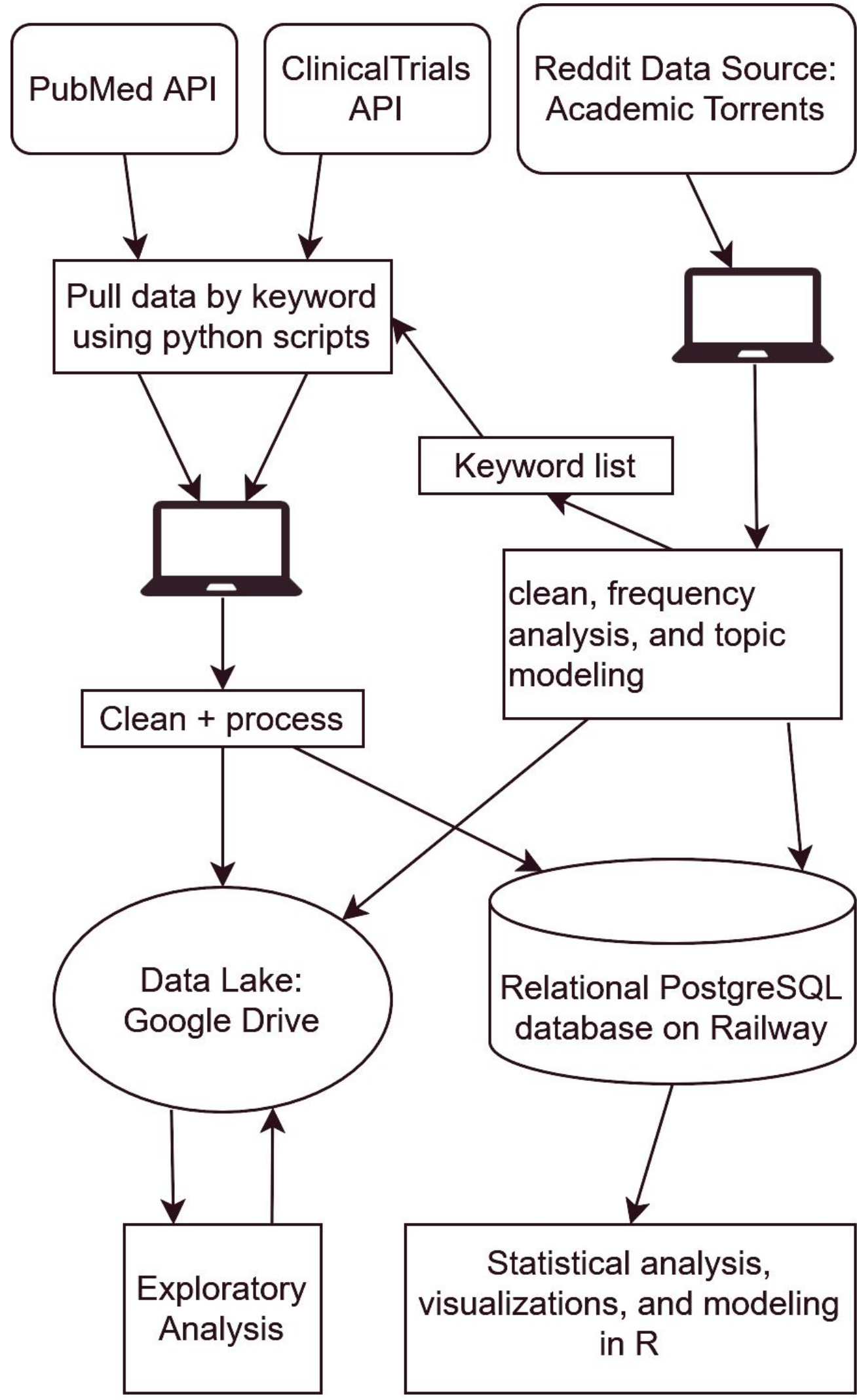
However, there is very little research comparing women’s lived experiences with published health literature. Our project aims to help close this gap.

Data, Engineering, and Ethics

Data
Reddit: 68,849 observations
• Date Range: 2013 -2025
PubMed: 2,235,186 observations
• Date Range: 1786 - 2026
ClinicalTrials: 101,542 observations
• Date Range: 1968 - 2026

Transformation
Observations were assigned a singular topic based on keywords. The following are the count distributions of each topic:

- Ethics**
- Personally identifiable information was removed from the Reddit data
 - Python API scraper packages are open source, rate limited.
 - Publication/trial data was pulled on topic modeled keywords. This could cause a gap of data.
 - Data is biased towards the U.S. and English-speaking countries
 - This project does not evaluate the intersectionality of race and gender within health publications.



Abstract

To identify **under-researched** areas of women’s health, we analyzed topics in a women’s health forum and their representation in research. We found that conditions affecting **both genders** are **~3.8x more researched** than those affecting just women, and several conditions had **10x less research than expected**, including but not limited to: **Bartholin cysts**, **yeast infections**, **hormonal acne**, and **vaginal health probiotics**.

Introduction

Women’s Health Include:

- Conditions of increased prevalence in women compared to men
- Pregnancy-related health disorders
- Gynecologic complications and cancers of specific tissues.^[1]

Conditions that disproportionately affect women remain **underfunded** and **understudied** relative to their prevalence, leading to a negative impact on women’s ability to understand their own health issues, receive diagnosis, treatment, and easily navigate healthcare systems.^[2]

A comparative analysis of experiential discourse from women and published medical understanding is crucial to map the gap in research.

Stakeholders

Primary stakeholder: **Women**

Who can make the change?

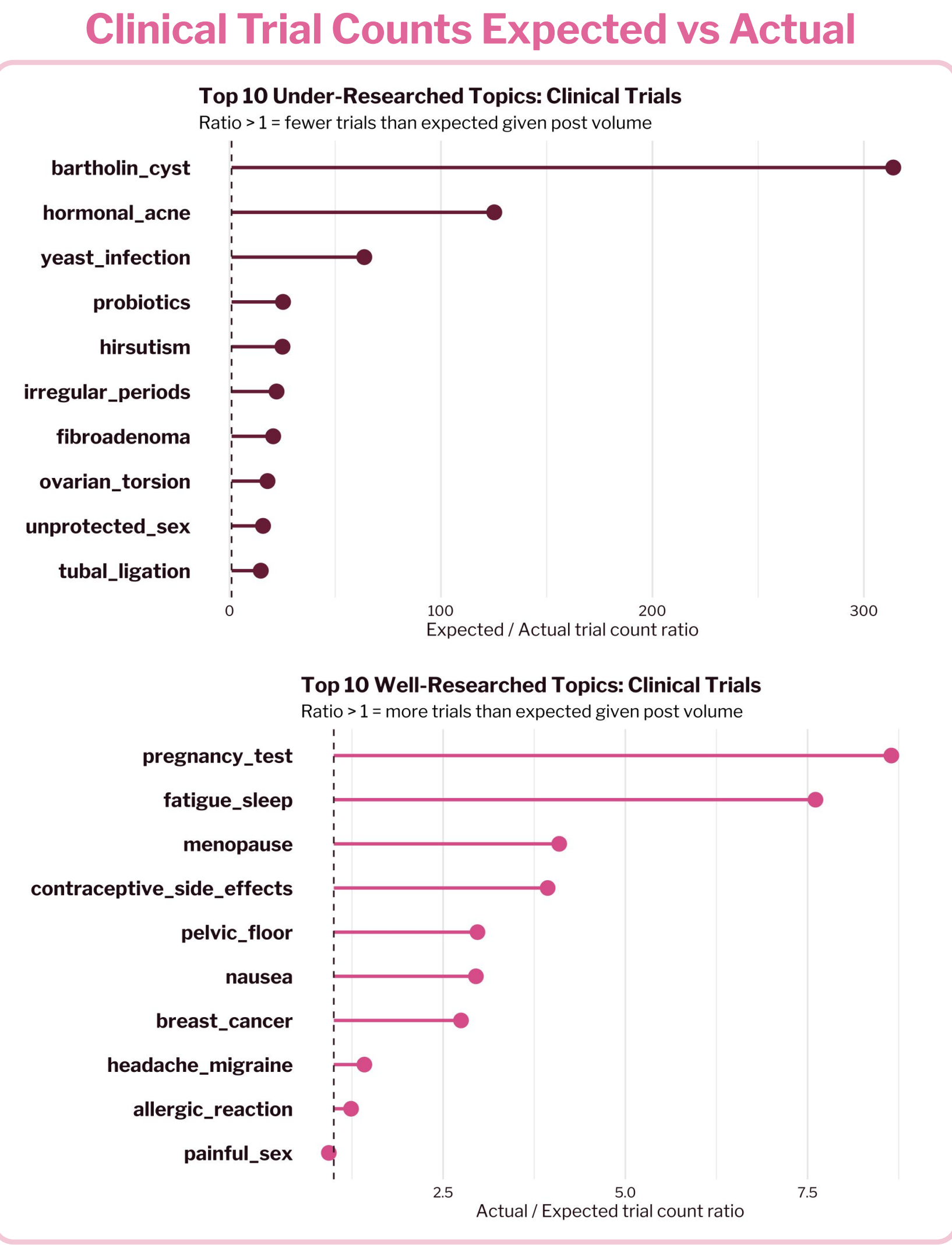
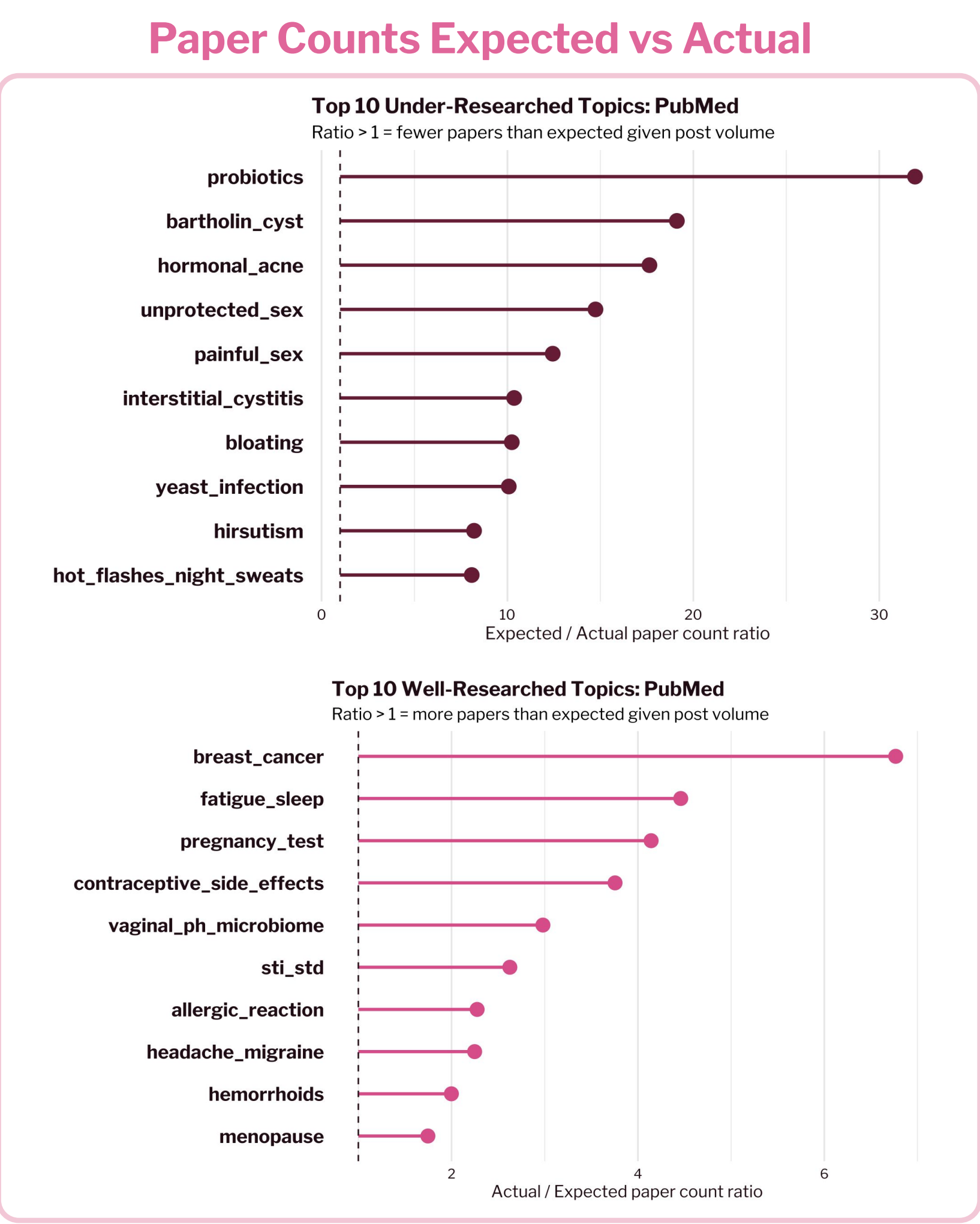
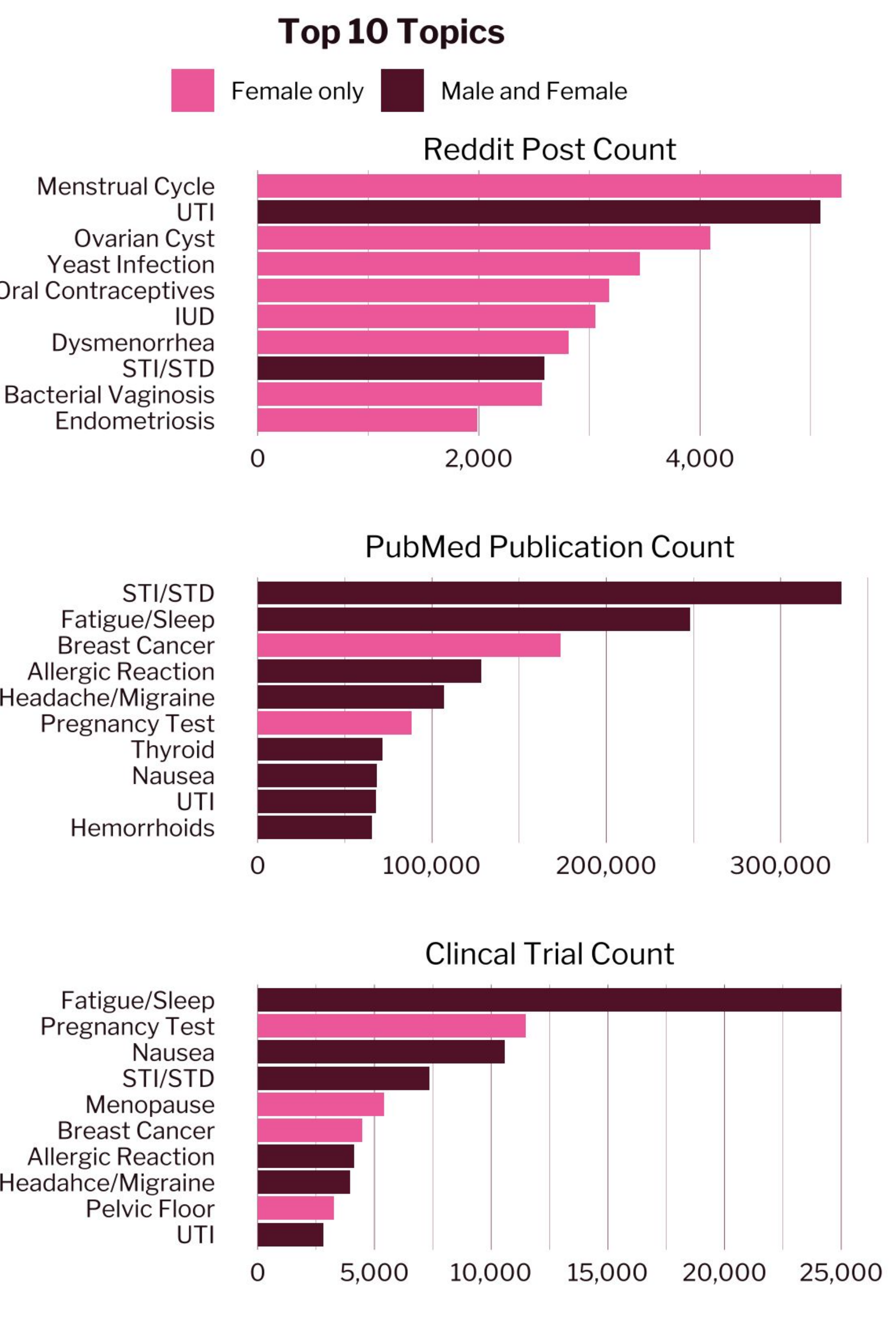
- Medical researchers
- Clinical trial designers
- Funders
- Public health policymakers
- Advocacy organizations

Results & Visualizations

Topics that affect **both genders** are **3.8x as researched** despite affecting just **2x** the population

A **1% increase** in post count means a **0.39% increase** in paper count and **0.42% increase** in trial count, showing the rate of discussion and research do not match.

Model: paper_count/clinical_trial_count ~ log(post_count) + both_sexes					
Predicted Value	Coefficient of post_count	P-value of post_count	Coefficient of both_sexes	P-value of both_sexes	Over-dispersion Parameter
Paper Count	0.3852	0.000981	1.3493	0.000243	1.3
Trial Count	0.4187	0.00170	1.3278	0.00156	1.69



Methods & Evaluations

Topic Modeling

We did **frequency analysis** to identify top keywords in discussion data and ran a G-test against expected frequencies to obtain a p-value. The top 50 keywords had p-values near zero, indicating statistical significance. Then, we used **BERTopic** to find topic clusters. Based on the results, we made a list of keywords to pull research data for.

Negative Binomial Regression Model

The final model is Negative Binomial Regression to account for overdispersion count data. A model was run with both ClinicalTrials.gov data and PubMed data as the response and Reddit posts as the predictor with the interaction of sex.

Reddit post count was log transformed within the model itself to account for the smaller values and right-skew.

Discussion, Limitations, and Conclusion

Discussion

Results found multiple topics that had 10x less research than expected and that topics affecting men and women get 3.8 as much research as women’s health topics. These findings contribute to data-driven evidence towards the known gap in women’s research.

Limitations

→ Bias in research and discussion data. All platforms are U.S. based and don’t generalize globally.

→ Limiting to one topic. Many publications, trials, and experiences are much more complex than one topic.

→ Limited data. Computing resources don’t allow for larger-scale intake and processing

Conclusion

The NIH and other funders should allocate more resources to topics like Bartholin cysts, hormonal acne, yeast infections, and more. Advocacy groups can use this data to support efforts towards topics that need it the most.

Future Work

Future work should include data from more publication sources from other countries, smaller journals, etc. With discussion data, surveys could be used to gain a more nuanced understanding of what women are talking about, what impacts them most, and what they struggle with, from a more diverse audience.

References & Reproducibility

GitHub Repository: <https://github.com/Addie-Shanti-Org/GenderDisparityinResearch>
References can be accessed via QR code or this link: <https://addie-shanti-org.github.io/qr/>

